

From dozens of SQL tables to digital biomarker features

Every raw table must be transformed before it can become a clinically interpretable feature family.

SQL tables tracked

44

Reviewed tables

10

Candidate features

69

Selected features

29

Calculated values

27

Each SensorDB table needs a dedicated extraction algorithm before it can contribute reliable model features.

This means table-by-table SQL logic, row interpretation, deduplication, privacy handling, feature calculation, and validation.

All SQL tables

• accelerometer	• network_traffic
✓ applications_foreground	• plugin_google_activity_recognition
• aware_device	• proximity
• aware_log	• rotation
• aware_studies	• screen
• barometer	• sensor_accelerometer
✓ battery	• sensor_barometer
• battery_charges	• sensor_bluetooth
• battery_discharges	• sensor_gravity
✓ bluetooth	• sensor_gyroscope
✓ calls	• sensor_light
• gravity	• sensor_linear_accelerometer
✓ gsm	• sensor_magnetometer
• gsm_neighbor	• sensor_proximity
• gyroscope	• sensor_rotation
✓ keyboard	• sensor_temperature
✓ light	• sensor_wifi
• linear_accelerometer	• significant
✓ locations	• telephony
• magnetometer	• timezone
✓ messages	• touch
• network	• wifi

Selected features already defined

29 selected features

applications_foreground	3 selected / 3 calculated
• app_foreground_event_count	
• unique_foreground_apps	
• app_use_diversity	
battery	2 selected / 0 calculated
• low_battery_event_count	
• charging_or_plugged_event_count	
bluetooth	2 selected / 2 calculated
• unique_bluetooth_addresses	
• bluetooth_address_diversity_ratio	
calls	5 selected / 5 calculated
• call_event_count	
• incoming_call_count	
• outgoing_call_count	
• missed_rejected_blocked_call_count	
• total_call_duration_seconds	
gsm	3 selected / 3 calculated
• unique_gsm_cell_count	
• unique_gsm_lac_count	
• gsm_cell_transition_count	
keyboard	6 selected / 6 calculated
• keyboard_median_inter_event_interval_ms	
• keyboard_inter_event_interval_iqr_ms	
• keyboard_long_pause_count_2s	
• keyboard_typing_burst_count	
• keyboard_median_word_completion_time_ms	
• keyboard_deletion_event_count	
light	4 selected / 4 calculated
• median_light_lux	
• percent_dark_samples	
• night_mean_light_lux	
• light_lux_iqr	
locations	2 selected / 2 calculated
• location_distinct_observation_count	
• location_total_distance_km	
messages	2 selected / 2 calculated
• message_distinct_event_count	
• outgoing_message_count	